

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior full-stack engineer with 5+ years shipping production systems end-to-end — Node.js + TypeScript + React on the application layer, Python data pipelines, Apache Kafka + NATS Streaming + gRPC microservices on Docker + Kubernetes for the infrastructure layer, GCP and AWS for cloud. Code at github.com/zac-09.

Background in social-impact technology: 18 months at CodeBits leading 4 engineers building legal-aid systems for NGOs in Uganda — case management for FIDA Uganda, paralegal database for LASPNET, and a USSD service over flaky 2G for users in rural areas. Now drawn to open-source climate and energy work as the natural extension. Async-first remote: 5+ years on US-based teams from Uganda (UTC+3, 1–2 hours ahead of CET).

CORE COMPETENCIES

Full-Stack End-to-End Shipping	Node.js / Express	TypeScript / JavaScript	React / React Native	Python (data + scripting)
Docker / Kubernetes	GCP / AWS	Apache Kafka / NATS / gRPC	Microservices	Zero-Downtime Migrations
Social-Impact Tech	Async Remote			

WORK EXPERIENCE

MTailor Sep 2022 – Present

Software Engineer · US Remote (Uganda, UTC+3)

- Led the end-to-end migration of **20+ production applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, reporting directly to the CTO.
- Built migration tooling in Python — read MongoDB, applied transformation logic, wrote Firestore + GCS.
- Implemented real-time two-way sync between MongoDB and Firestore using Node.js and Google Pub/Sub.
- Built a 3D visualization feature using video overlay and ffmpeg, increasing buyer conversion.
- Saved **~\$5,000/month** by migrating off AWS.

CodeBits Jan 2020 – Jul 2021

Software Engineer · Lead of 4 · Social-Impact Tech for NGOs

- Led a team of 4 building legal-tech systems for NGOs serving vulnerable populations in Uganda.
- Built a case management system for **FIDA Uganda** and a paralegal database for **LASPNET Uganda**.
- Built a **USSD legal-aid service over flaky 2G** — Node.js backend with retry-aware session state machine and per-region latency instrumentation; worked on networks where REST timed out.
- Architected a microservices backend with **Apache Kafka, NATS Streaming, gRPC, Docker, and Kubernetes** — independent deploys, 4 NGO tenants on shared infra.
- Built a cross-platform mobile app for LASPNET using React Native + Expo with Firebase push notifications.

Dr Wealth Aug 2021 – Jan 2022

Software Engineer · Singapore Remote

- Optimized a PWA delivering real-time stock prices from a Firebase backend.
- Built ad-hoc analytical jobs querying **2M+ Firestore records**, keeping prices current from Morningstar APIs.
- Extended backend APIs using Firebase Cloud Functions and Express.js on Heroku.

Mind2matter.co Jan 2022 – Jul 2022

Software Engineer · US Remote

- Built backends for DeFi applications using Web3 and Node.js.
- Delivered React UIs from Figma designs under tight client timelines.

EDUCATION

B.Sc. Computer Engineering — **Makerere University** Aug 2018 – Sep 2022

SKILLS

Proficient: Node.js, Express, TypeScript, JavaScript, React, React Native, Python (scripting + data pipelines), Firebase/GCP, MongoDB, SQL, Docker, Kubernetes, Apache Kafka, NATS Streaming, gRPC, Linux, Git, GitHub Actions

Intermediate: Go, Rust, AWS, Web3

Open to ramp: Flask (Express-equivalent mental model), OpenTofu/Terraform, Nix/NixOS, Rescript

Learning: PyPSA-Eur source — happy to ship a small PR before/during interview process

Languages: English (native-level), Kiswahili